



A1	CRANBROOK	✓	✗
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Find the equation of the line perpendicular to the line $y = 2x - 3$ and passing

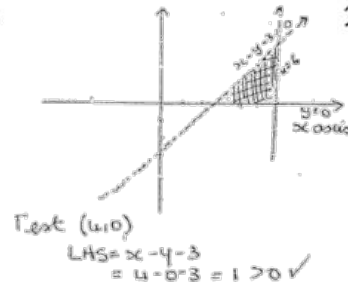
through the point (1,1). 2

$y = 2x - 3$
 $m = 2$
 $\therefore$ line perpendicular
 will have $m = -\frac{1}{2}$
 $y - 1 = -\frac{1}{2}(x - 1)$
 $2(y - 1) = -(x - 1)$
 $2y - 2 = -x + 1$
 $x + 2y - 3 = 0$

A2	KAMBALA	✓	✗
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Shade the region in the Cartesian plane for which the following inequalities hold simultaneously:

$x - y - 3 > 0$
 $x \leq 6$
 $y \geq 0$



A3	ST CATHERINE	✓	✗
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Find the equation of the line which passes through the point (1,4) and is perpendicular to the line $3x - 2y - 4 = 0$, giving your answer in general form. 2

$3x - 2y - 4 = 0$
 $m = \frac{3}{2}$

$\therefore m_{\perp} = -\frac{2}{3}$
 $\therefore y - 4 = -\frac{2}{3}(x - 1)$
 $2x + 3y - 14 = 0$

A4	PLC	✓	✗
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The line k passes through the points $A(2,1)$ and $B(1,3)$.

- Show that the equation of line k is $2x + y - 5 = 0$. 2
- Show that the point $(6, -7)$ lies on the line k . 1
- Find the co-ordinates of C such that A is the midpoint of CB . 2
- Show that $AB = OA$, where O is the origin. 2
- Find the equation of the line perpendicular to k , passing through the origin. 1
- Find the equation of the circle that passes through the points O , C and B . 2

a) Sub in point $A(2,1)$
 $LHS = 2(2) + 1 - 5$
 $= 0 = RHS$ $\therefore$ true (2)
 Sub in point $B(1,3)$
 $LHS = 2(1) + 3 - 5 = 0$

b) Sub in $x = 6$ and $y = -7$
 $2x + y - 5 = 0$
 $LHS = 2(6) + (-7) - 5$
 $= 0 = RHS$ (1)
 $\therefore (6, -7)$ lies on the line

c) $2 = \frac{x+1}{2}$ $1 = \frac{y+3}{2}$ (2)
 $4 = x + 1$ $2 = y + 3$
 $x = 3$ $y = -1$
 $C(3, -1)$

d) $AB = \sqrt{(2-1)^2 + (1-3)^2}$
 $= \sqrt{5}$
 $OA = \sqrt{(2-0)^2 + (1-0)^2}$
 $= \sqrt{5}$
 $\therefore AB = OA = \sqrt{5}$

e) $m_1(k) = -2$
 $\therefore m_2 = \frac{1}{2}$
 Point $(0,0)$
 $y - 0 = \frac{1}{2}(x - 0)$
 $y = \frac{1}{2}x$
 $x + 2y = 0$

f) $r = \sqrt{5}$ $C(2,1)$
 $(x-2)^2 + (y-1)^2 = 5$



A5	RANDWICK GIRLS	☒	☐
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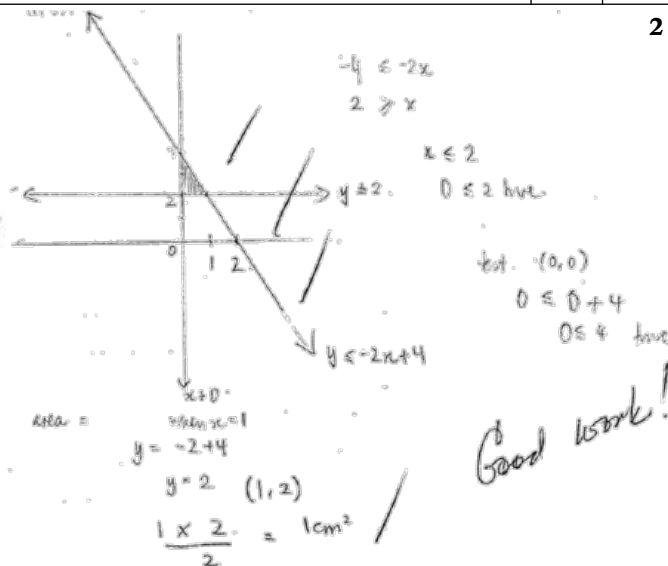
Sketch the region given by:

$$y \geq 2$$

$$y \leq -2x + 4$$

$$x \geq 0$$

What is the area of this shaded region?



2

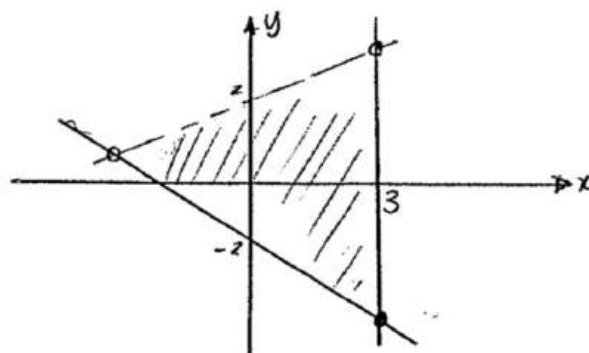
A6	SYDNEY BOYS	☒	☐
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Sketch on the number plane the region in which all three of the following inequalities are satisfied:

$$x - y + 2 > 0$$

$$2x + y \geq -2$$

$$x - 3 \leq 0$$



2

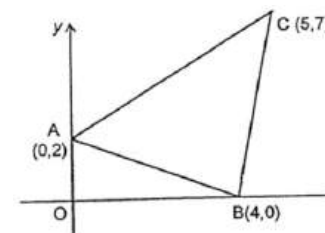
A7	SYDNEY BOYS	☒	☐
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The diagram shows the points $A(0, 2)$, $B(4, 0)$, and $C(5, 7)$.

6

Copy the diagram onto your work sheet.

- Find the co-ordinates of M , the mid-point of AB .
- Show that the gradient of AB is $-\frac{1}{2}$.
- Find the equation of the perpendicular bisector of AB .
- Show that the perpendicular bisector of AB passes through C .
- What type of triangle is ABC ? (Give a reason for your answer.)



$$(a) \quad M: x = \frac{4+0}{2}, y = \frac{2+0}{2}$$

$$= 2 \quad = 1$$

$$\therefore M(2, 1)$$

$$(b) \quad m_{AB} = \frac{0-2}{4-0}$$

$$= -\frac{1}{2}$$

$$\therefore m_{AB} = -\frac{1}{2}$$

$$(c) \quad m_1 \times m_2 = -1$$

$$-\frac{1}{2} \times m_2 = -1$$

$$m_2 = 2$$

$$\text{Using } y - y_1 = m(x - x_1), \text{ where } x = 2, y = 1, m_2 = 2$$

$$y - 1 = 2(x - 2)$$

$$\therefore 2x - y - 3 = 0$$

$$(d) \quad \text{Using } 2x - y - 3 = 0, \text{ where } x = 5, y = 7$$

$$2 \times 5 - 7 - 3 = 0$$

$$\therefore LHS = RHS$$

$$(e) \quad \text{In } \triangle ABC$$

$$d_{AC} = d_{BC} = \sqrt{50}$$

$$d_{AB} = \sqrt{20}$$

$$\therefore \triangle ABC \text{ is isosceles}$$



A8	SYDNEY BOYS	☒	☒
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The diagram above shows the points $A(-2,1)$, $B(3,5)$, and $C(4,-1)$.

Copy the diagram to your answer booklet.

- (i) Find the equation of the line through the points A and B .
- (ii) Write the equation of the line through C perpendicular to AB .

i). $A(-2,1)$, $B(3,5)$, $C(4,-1)$

$$m_{AB} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5-1}{3-(-2)} = \frac{4}{5}$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{4}{5}(x + 2)$$

$$y - 1 = \frac{4}{5}x + \frac{8}{5}$$

$$5y - 5 = 4x + 8$$

$$0 = 4x - 5y + 13$$

ii). $m_1 = \frac{4}{5}$

$$m_2 = -\frac{5}{4} \text{ (lines are perpendicular)}$$

$$y - y_1 = m(x - x_1)$$

$$y + 1 = -\frac{5}{4}(x - 4)$$

$$y + 1 = -\frac{5}{4}x + 5$$

$$4y + 4 = -5x + 20$$

$$5x + 4y - 16 = 0$$

A9	SYDNEY GIRLS	☒	☒
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The vertices of the triangle ABC are the points $A(0,0)$, $B(0,2)$ and $C(3,-1)$.

- i) Draw a sketch diagram of the triangle. (1)
- ii) The point K on BC is such that AK is perpendicular to BC . Find the coordinates of K , and show the point K on your diagram. (2)
- iii) Find the area of the triangle ABC . (1)

$$m_{BC} = \frac{-1-2}{3-0} = -1$$

BC

$$y - 2 = -1(x - 0)$$

$$y = -x + 2$$

$$y = 2$$

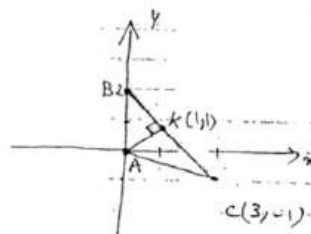
$$x = -x + 2$$

$$2x = 2$$

$$x = 1$$

$$y = 1$$

$$K(1,1)$$



ii). $AK = \sqrt{(1-0)^2 + (1-0)^2} = \sqrt{2}$

$$BC = \sqrt{(3-0)^2 + (-1-2)^2} = \sqrt{18}$$

$$Area = \frac{1}{2} \times \sqrt{2} \times \sqrt{18}$$

$$= \frac{1}{2} \times \sqrt{36}$$

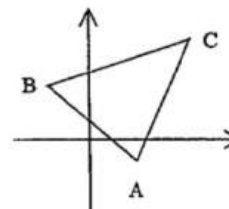
$$= \frac{1}{2} \times 6 = 3$$

$$Area = 3$$

$$Area = 3$$

A10	SCEGGS	☒	☒
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The three points $A(2,-1)$, $B(-2,3)$ and $C(4,5)$ form a triangle.



- (i) Find the coordinates of P , the midpoint of BC . 1
- (ii) Show that Q , the midpoint of AB lies on the y axis. 1
- (iii) Find the equation of PQ . 2
- (iv) Show that $PQ \parallel AC$. 2
- (v) Show that AC is twice the length of PQ . 2

i) $P(1,4)$ ✓

ii) $Q(0,1)$ which lies on the y axis ✓

iii) $m_{PQ} = \frac{4-1}{1-0} = 3$ ✓

$$y - y_1 = m(x - x_1)$$

$$y - 4 = 3(x - 1)$$

$$y = 3x + 1$$

iv) $m_{AC} = \frac{5-(-1)}{4-2} = 3$ ✓

$\therefore$ Since $m_{AC} = m_{PQ}$,

$$AC \parallel PQ.$$

v) $AC = \sqrt{(4-2)^2 + (5-(-1))^2} = \sqrt{40} = 2\sqrt{10}$ ✓

$$PQ = \sqrt{(1-0)^2 + (4-1)^2} = \sqrt{10}$$

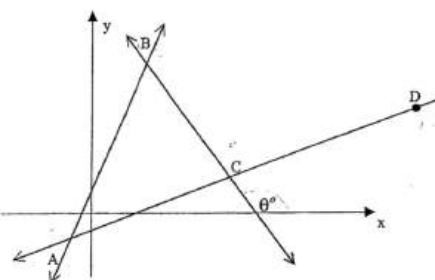
$\therefore AC = 2 \times PQ.$



A11	SCOTS	☒	☐
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The diagram shows the points A(-2,-2), B(2,5), C(4,1) and D(x,4).

- Calculate the length of AB leaving your answer in surd form. (1mark)
- Show that the gradient of the line BC is -2. (1mark)
- Find the angle θ that the line BC makes with the positive direction of the x-axis. Write your answer to the nearest minute. (1mark)
- D is a point which lies on the line AC. If C is the midpoint of interval AD, show that the co-ordinates of point D are (10,4). (1mark)
- Show that the line BC is perpendicular to the line AD. (2marks)
- Given that the gradient of the line AB is $\frac{7}{4}$.
Find the equation of line AB. (2marks)



$$\begin{aligned}
 a) \quad AB &= \sqrt{4^2 + 7^2} = \sqrt{65} \\
 b) \quad m_{BC} &= \frac{(5) - (1)}{(2) - (4)} = \frac{4}{-2} = -2 \\
 c) \quad \tan \theta &= -2 \quad \therefore \theta = 180^\circ - 63^\circ 26' \\
 &\quad \therefore \theta = 116^\circ 34' \\
 d) \quad \text{Let } D &= (x, y) \therefore C = \left(\frac{x-2}{2}, \frac{y-2}{2} \right) = (4, 1) \\
 \frac{x-2}{2} &= 4 \rightarrow x = 10 \\
 \frac{y-2}{2} &= 1 \rightarrow y = 4 \\
 e) \quad m_{AD} &= \frac{(4) - (-2)}{(10) - (-2)} = \frac{6}{12} = \frac{1}{2} \\
 &\quad \text{Since } m_{BC} \times m_{AD} = -2 \times \frac{1}{2} = -1 \\
 &\quad \therefore AD \perp BC \\
 f) \quad y - 5 &= \frac{7}{4}(x - 2) \\
 4y - 20 &= 7x - 14 \\
 \therefore 7x - 4y + 6 &= 0
 \end{aligned}$$

A12	SYDNEY GIRLS	☒	☐
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Find the equation of the straight line which passes through the point of intersection of $3x - y + 4 = 0$ and $x + 2y + 3 = 0$ and is parallel to $3y - x + 2 = 0$.

$$\begin{aligned}
 3x - y + 4 &= 0 \rightarrow y = 3x + 4 \\
 x + 2y + 3 &= 0 \\
 x + 2(3x + 4) + 3 &= 0 \\
 7x + 11 &= 0 \\
 \therefore x &= -\frac{11}{7} \quad y = -\frac{6}{7} \\
 \text{Parallel to } 3y - x + 2 &= 0 \\
 \therefore m_1 &= m_2 = \frac{1}{3}
 \end{aligned}$$

$$\begin{aligned}
 \text{Eqn: } y + \frac{5}{7} &= \frac{1}{3}\left(x + \frac{11}{7}\right) \\
 3y + \frac{15}{7} &= x + \frac{11}{7} \\
 21y + 15 &= 7x + 11 \\
 \therefore 7x - 21y - 4 &= 0
 \end{aligned}$$

A13	SYDNEY GIRLS	☒	☐
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Find the equation of the straight line through (3,-2) that passes through the intersection of the lines $5x + 2y - 13 = 0$ and $x - 3y + 11 = 0$.

$$\begin{aligned}
 5x + 2y - 13 &= 0 \\
 x - 3y + 11 &= 0 \rightarrow x = 3y - 11 \\
 5(3y - 11) + 2y - 13 &= 0 \\
 15y - 55 + 2y - 13 &= 0 \\
 17y - 68 &= 0 \\
 17y &= 68 \\
 y &= 4 \\
 \therefore x &= 1
 \end{aligned}$$

$$\begin{aligned}
 y - 4 &= \left(\frac{4+2}{1-3} \right) (x - 1) \\
 y - 4 &= -3(x - 1) \\
 y - 4 &= -3x + 3 \\
 3x + y - 7 &= 0
 \end{aligned}$$



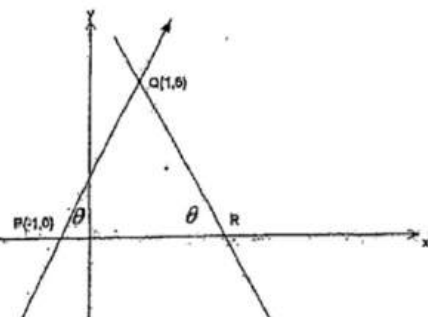
A14

SYDNEY GIRLS



In the diagram above P and Q have co-ordinates $(-1, 0)$ and $(1, 6)$ and $\angle QPR = \angle QRP = \theta$.

- Find the co-ordinates of the midpoint of PQ.
- Show that PQ has equation $y = 3x + 3$.
- Show that $\tan \theta = 3$ hence find θ correct to the nearest minute
- Find the co-ordinates of R.
- Show that the gradient of QR is -3
- Show that the equation of QR is $3x + y - 9 = 0$
- On your diagram, shade the region satisfying both the inequalities:
 $y \leq 3x + 3$ and $3x + y - 9 \geq 0$



a) P $(-1, 0)$ and Q $(1, 6)$

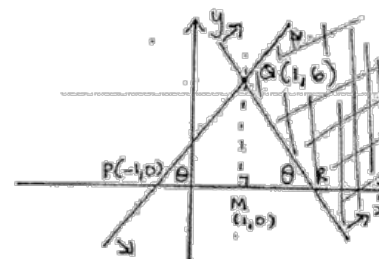
$$\begin{aligned} \text{distance } PQ &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(1 - (-1))^2 + (6 - 0)^2} \\ &= \sqrt{(-6)^2 + (-2)^2} \\ &= \sqrt{36 + 4} = \sqrt{40} \\ &= 2\sqrt{10} \end{aligned}$$

(2)

b) Area = $\frac{1}{2} \times \text{base} \times \text{height}$
 $= \frac{1}{2} \times (9 - 2) \times 13 \sin 29^\circ$
 $= 28.99 \text{ m}^2$

(3)

c)



i) Midpoint PQ $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
 $= \left(\frac{-1 + 1}{2}, \frac{0 + 6}{2} \right)$
 $= (0, 3)$

(ii) Gradient PQ = $\frac{y_2 - y_1}{x_2 - x_1}$
 $= \frac{6 - 0}{1 - (-1)} = \frac{6}{2}$
 $m = 3$

Equation $y - y_1 = m(x - x_1)$
 $y - 6 = 3(x - 1)$

$y - b = mx + c$
 $y = 3x + 3$

(3)

(iii) since gradient $m = 3$

$\therefore \tan \theta = m$

$\tan \theta = 3$

hence $\theta = 71^\circ 34'$

(2)

(iv) Since ΔPQR is isosceles, draw PM from Q.

$\therefore M$ has co-ordinates $(1, 0)$

$\therefore PM = MR$

$\therefore R$ has co-ordinates $(3, 0)$

(2)

v) Gradient QR = $\frac{y_2 - y_1}{x_2 - x_1}$
 $= \frac{0 - 6}{3 - 1} = \frac{-6}{2}$
 $= -3$

(2)

vi) Equation QR

$m = -3, Q(1, 6)$

$y - y_1 = m(x - x_1)$

$y - 6 = -3(x - 1)$

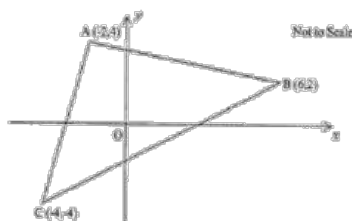
$y - 6 = -3x + 3$

$3x + y - 9 = 0$

(3)

A15	INDEPENDENT	☒	☐
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The diagram below shows the points A (-2, 4), B (6, 2) and C (-4, -4). Copy or trace the diagram onto your worksheet.



- Calculate the length of the interval BC.
- Find the gradient of BC.
- Find the coordinates of M, the midpoint of BC.
- Show that the equation of l , the perpendicular bisector of BC, is $5x + 3y - 2 = 0$.
- Show that l passes through A.
- Hence or otherwise find the area of triangle ABC.

1
1
1
2
1
2

$$(i) BC = \sqrt{(-2+4)^2 + (2+4)^2}$$

$$= \sqrt{136}$$

$$= 2\sqrt{34}$$

$$(ii) m = \frac{2+4}{6+4} = \frac{3}{5}$$

$$(iii) M = \left(\frac{-2+6}{2}, \frac{4+2}{2} \right)$$

$$= (2, 1)$$

$$(iv) y+1 = \frac{-5}{3}(x-1)$$

$$3y+3 = -5x+5$$

$$5x+3y-2=0$$

$$(v) 5(-2) + 3(4) - 2$$

$$= -10 + 12 - 2$$

$$= 0$$

$$(vi) AM = \sqrt{(-2-2)^2 + (4-1)^2}$$

$$= \sqrt{34}$$

$$\therefore \text{Area} = \frac{1}{2} \cdot 2\sqrt{34} \cdot \sqrt{34}$$

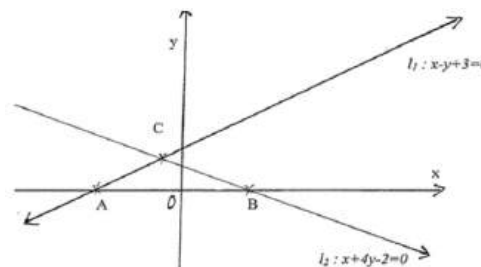
$$= 34 \text{ units}^2$$

A16	ST CATHERINE	☒	☐
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In the diagram, l_1 and l_2 with equations $x-y+3=0$ and $x+4y-2=0$, meet at C. The lines cut the x-axis at A and B respectively.

- Find the coordinates of the points A and B.
- Solve simultaneously to find the point C.
- Find the angle of inclination of the line l_1 to the positive x-axis.

2
3
2



$$3. a) A: \text{sub } y=0:$$

$$x - 0 + 3 = 0$$

$$\therefore x = -3 \therefore A(-3, 0)$$

$$B: \text{sub } y=0$$

$$x + 0 - 2 = 0$$

$$x = 2 \therefore B(2, 0)$$

$$b) x - y + 3 = 0 \quad (1)$$

$$x + 4y - 2 = 0 \quad (2)$$

$$(1) - (2) \quad -5y + 5 = 0$$

$$\therefore 5y = 5$$

$$y = 1$$

$$\text{sub } y=1 \text{ into } (1)$$

$$x - 1 + 3 = 0$$

$$\therefore x + 2 = 0$$

$$x = -2 \therefore C(-2, 1)$$

$$c) x - y + 3 = 0 \Rightarrow y = x + 3$$

$$\therefore m = 1 \therefore \tan \theta = m$$

$$\therefore \tan \theta = 1$$

$$\therefore \theta = 45^\circ$$

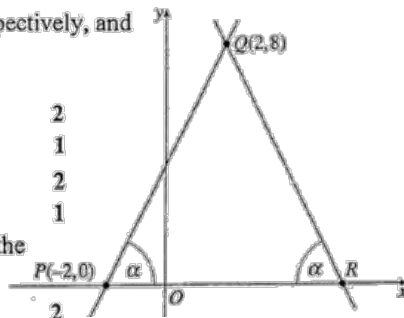
$$\therefore \text{angle of inclination is } 45^\circ$$



A17	SYDNEY GIRLS	☒	☐
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In the diagram, P and Q have coordinates $(-2,0)$ and $(2,8)$ respectively, and $\angle QPR = \angle QRP = \alpha$.

- Find the gradient of PQ .
Hence, or otherwise show that $\tan \alpha = 2$
- Show that the gradient of QR is -2 .
- Show that the equation of QR is $2x + y - 12 = 0$.
- Find the co-ordinates of R .
- On your diagram, shade in the region satisfying both the inequalities:
 $y \leq 2x + 4$ and $2x + y - 12 \geq 0$



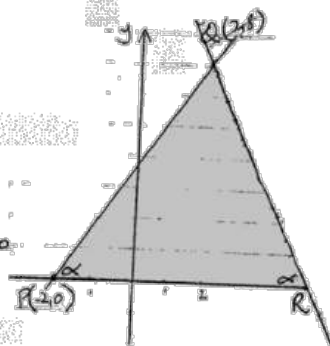
$$\begin{aligned} \text{ii) } m_{PQ} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{8 - 0}{2 - (-2)} \\ &= \frac{8}{4} \\ &= 2 \checkmark \end{aligned}$$

$$\begin{aligned} \text{using } m &= \tan \alpha \\ \therefore \tan \alpha &= 2 \checkmark \end{aligned}$$

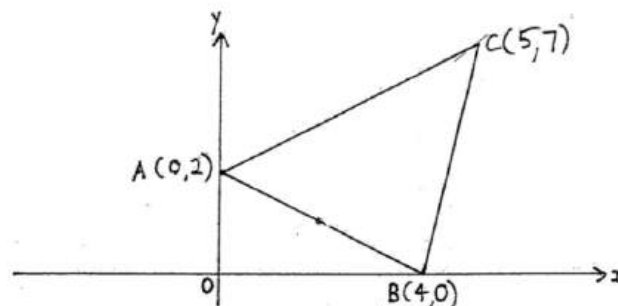
$$\begin{aligned} \text{iv) } \angle QRP &= 180^\circ - \angle QPR - \alpha \\ \therefore \text{gradient of } QR &= \tan \angle QRP \\ &= \tan(180^\circ - \alpha) \\ &= -\tan \alpha \\ &= -2 \end{aligned}$$

$$\begin{aligned} \text{ii) } m_{QR} &= -2 \\ Q(2,8) \\ y - y_1 &= m(x - x_1) \\ y - 8 &= -2(x - 2) \\ y - 8 &= -2x + 4 \\ 2x + y - 8 - 4 &= 0 \\ 2x + y - 12 &= 0 \end{aligned}$$

$$\begin{aligned} \text{iv) } R \text{ lies on line } QR \\ \text{and the } x\text{-axis} \\ \therefore \text{the } y\text{-coordinate is } 0 \\ \therefore \text{when } y = 0 \\ 2x + 0 - 12 = 0 \\ 2x = 12 \\ x = 6 \\ \therefore R(6,0) \end{aligned}$$



A18	SYDNEY TECH	☒	☐
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- Find the gradient of AB 1
- Find the coordinates of D , the midpoint of AB 1
- Find the equation of the line passing through D and perpendicular to AB 2
- Show that C lies on this line 1
- Find the lengths of AB and CD in surd form. 2
- Find the area of the quadrilateral $ACBO$ 2

$$\begin{aligned} \text{iii) } m &= 2 \quad (2,1) \\ y - 1 &= 2(x - 2) \quad \text{①} \\ y - 1 &= 2x - 4 \\ y &= 2x - 3 \text{ or } 2x - y - 3 = 0 \end{aligned}$$

$$\text{b) ci) } m_{AB} = \frac{-1}{2} \quad \text{①}$$

$$\text{ii) } D(2,1) \quad \text{①}$$

$$\begin{aligned} \text{ci) } 7 &= 2 \times 5 - 3 \\ 7 &= 10 - 3 \checkmark \quad \text{①} \end{aligned}$$

$$\begin{aligned} \text{iv) } d_{AB} &= \sqrt{(4-0)^2 + (0-2)^2} \\ &= \sqrt{20} \text{ or } 2\sqrt{5} \quad \text{①} \end{aligned}$$

$$\begin{aligned} d_{CD} &= \sqrt{(5-2)^2 + (7-1)^2} \\ &= \sqrt{9 + 36} \\ &= \sqrt{45} \text{ or } 3\sqrt{5} \quad \text{①} \end{aligned}$$

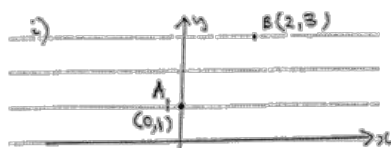
$$\begin{aligned} \text{vi) Area} &= \frac{1}{2} \times 4 \times 2 + \frac{1}{2} \times 2\sqrt{5} \times 3\sqrt{5} \\ &= 4 + 15 \\ &= 19 \text{ units}^2 \quad \text{①} \end{aligned}$$

A19	SYDNEY TECH	☒	☐
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Let A and B be the points (0, 1) and (2, 3) respectively

- | | |
|---|---|
| (i) Find the coordinates of the midpoint of AB | 1 |
| (ii) Find the slope of the line AB | 1 |
| (iii) Find the equation of the perpendicular bisector of AB | 3 |
| (iv) The point P lies on the line $y = 2x - 9$ and is equidistant from A and B. | 2 |

Using your answer to part (iii) or otherwise, find the coordinates of P.



i) midpt $(1, 2)$ ①

ii) $m_{AB} = \frac{3-1}{2-0} = \frac{2}{2} = 1$ ①

iii) perp bisector ① for reciprocal
 $y - 2 = -1(x - 1)$ ① for point
 $y - 2 = -x + 1$ gradient
 $x + y - 3 = 0$ formula ①

iv) $x + y - 3 = 0$ ① solve
 $2x - y - 9 = 0$ ② simultaneously
 ① + ② gives
 $3x - 12 = 0$
 $x = 4$ ③ sub into ①
 $4 + y - 3 = 0$
 $y = -1$ ④
 P (4, -1)